

Build Your Own STEREO AMPLIFIER Using TDA2822

RAJ K. GORKHALI

Here is a simple stereo amplifier that can be used to amplify audio from mobile devices. IC TDA2822M used in the circuit comes in an 8-pin DIP package and is readily available in the market.

Circuit and working

Circuit diagram of the stereo audio amplifier is shown in Fig. 1. In addition to IC TDA2822M (IC1), it uses four capacitors (C1 through C4), two

potmeters (VR1 and VR2) and two speakers. The circuit works off 3V DC. However, you can use up to 6V DC.

The circuit diagram is straightforward. The left speaker (LS1) is connected to output pin 1 of IC1 through electrolytic capacitor C3. The right speaker (LS2) is connected to output pin 3 through electrolytic capacitor C4. Pins 5 and 8 are tied together and connected to ground via electrolytic capacitor C1. Pins 7 and 6 are input pins and connected

to potmeters VR1 and VR2, respectively. VR1 and VR2 work as left and right-channel volume controls, respectively. Pin 2 is connected to 3V DC supply and pin 4 is connected to ground. Electrolytic capacitor C2 connected across 3V and ground works

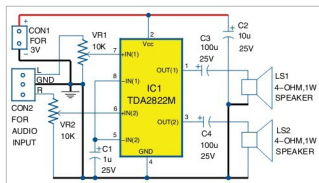


Fig. 1: Circuit of the stereo audio amplifier using TDA2822M

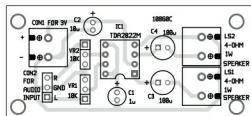


Fig. 2: Actual-size PCB layout of stereo audio amplifier using TDA2822M

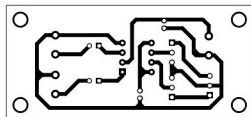


Fig. 3: Components layout for the PCB

as a filter capacitor.

Connect audio signal input from your mobile to CON2, and connect 3V DC supply across CON1. Both speakers produce the same audio output after amplifying the signal from IC1.

Construction and testing

An actual-size PCB layout of the stereo audio amplifier using TDA2822M is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit, enclose it in a suitable box. Fix potmeters VR1 and VR2 on the front panel of the box for left and right volume

PARTS LIST

Semiconductors:	
IC1	- TDA2822M dual power amplifier
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
VR1, VR2	- 10-kilo-ohm potentiometer
Capacitors:	
C1	- 1 μ F, 25V electrolytic
C2	- 10 μ F, 25V electrolytic
C3, C4	- 100 μ F, 25V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector
CON2	- 3-pin connector
LS1, LS2	- 4-ohm, 1W speaker
	- 3V battery (two 1.5V pencil cells)
	- 2-pin terminal connectors for LS1 and LS2
	- 8-pin IC socket

control, respectively. On the PCB, use an 8-pin IC socket for TDA2822M so that you can easily remove the IC during troubleshooting.

Calibration and adjustment

For calibration, take a 3V battery; you can use two 1.5V pencil cells and put them in a battery holder. Now, connect speakers to output connectors LS1 and LS2 marked on the PCB. Set potmeters VR1 and VR2 to their middle position. Take a metal screwdriver and gently touch on input pin 6 or 7 of IC1. You should hear a humming sound from left and right speakers when the screwdriver is touched at left and right inputs, respectively. Slowly adjust VR1 clockwise until you hear a loud humming sound from the left speaker. Similarly, adjust VR2 for the right speaker. Now your stereo amplifier is working and ready for use! **EFY**



Raj K. Gorkhali is a regular contributor to EFY who takes keen interest in designing electronics circuits